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One-Dimensional Acoustic Waves in Retarding Structures with Propagation Velocity Tending to Zero

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Abstract—A retarding structure that allows the effective admittance of a tube wall to increase smoothly along the tube axis is considered. The sound velocity gradually decreases along a finite segment of the tube and finally vanishes at some cross section. The time of the sound propagation along this segment is infinitely long. A wave incident on the input cross section cannot reach the other end of the tube within a finite time, and, hence, it is not reflected from it. The wave is completely absorbed, the absorption being caused by the energy accumulation in the cross section where the velocity of sound vanishes, rather than by the energy transformation to heat, as in common sound absorbers. A differential equation is obtained to describe the sound propagation in a one-dimensional waveguide with a varying cross section and varying acoustic admittance of the walls. The solutions to this equation are analyzed in the WKB approximation. An exact solution is determined for the case of some specific functions describing the variations of the cross section and admittance. Calculated results for the input admittance of the waveguide are presented. A possible similarity to the problem of shear waves in sea sediments is pointed out. © 2002 MAIK “Nauka/Interperiodica”.

GENERALIZED WEBSTER EQUATION

The velocity of sound propagation in a narrow (in comparison with the wavelength) tube depends on the acoustic admittance of the tube wall [1]. The elastic-type wall admittance reduces the velocity of propagation. In what follows, we investigate longitudinal sound waves in a narrow waveguide of varying cross section with varying acoustic admittance of the walls. The waveguide provides a smooth decrease in the velocity of wave propagation, which, however, occurs rapidly enough for the propagation velocity to vanish within a finite distance. The possibility of creating specially tapered edges of plates and bars to eliminate the reflection of bending waves was considered earlier [2, 3]. Two conditions determine the choice of the tapering rate. On the one hand, the cross section should vary smoothly to cause no reflection. On the other hand, the cross section should vary rapidly enough for the velocity of a bending wave to vanish within the tapered segment of a finite length.

In the case of a narrow, axially symmetric waveguide of varying cross section and with varying wall admittance, the wave equation is derived as follows. We direct the x axis along the symmetry axis and denote by $S = S(x)$ the area of the waveguide cross section, by $r = r(x)$ the waveguide radius, and by $Y = Y(x)$ the admittance of the walls (Fig. 1). The Euler equation remains intact:

$$-\frac{\partial p}{\partial x} = \rho_0 \frac{\partial v}{\partial t}, \quad (1)$$

where v is the projection of the velocity on the x axis, p is the pressure, and ρ_0 is the density of the medium. The equation of continuity can be obtained from the conservation of matter in a thin layer of thickness dx (see Fig. 1)

$$d(\rho_0 S v) + (2\pi r v_{\perp} \rho_0) dx + \frac{\partial \rho}{\partial t} S dx = 0,$$

where $v_{\perp}$ is the projection of the velocity of liquid near the walls on the perpendicular to the symmetry axis and ρ is the acoustic component of the density. Substituting $v_{\perp} = Yp$, we obtain

$$-\frac{1}{\rho_0} \frac{\partial \rho}{\partial t} = v (\ln S)'_x + \frac{\partial v}{\partial x} + \frac{2Yp}{r}. \quad (2)$$

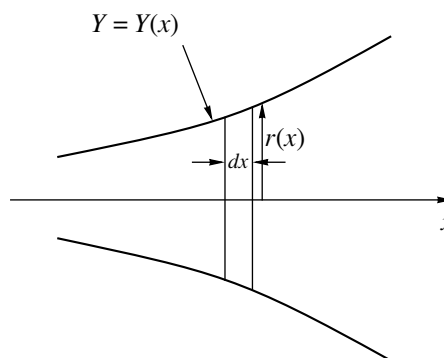


Fig. 1. Derivation of the generalized Webster equation. The waveguide with a varying cross section and varying wall admittance.

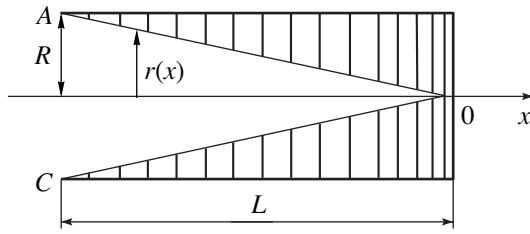


Fig. 2. Model of an acoustic black hole. The tube with rings whose inner radius $r(x)$ decreases to zero.

Taking into account the equality $\frac{\partial \rho}{\partial t} = \frac{1}{c^2} \frac{\partial p}{\partial t}$ and eliminating velocity v from Eqs. (1) and (2), we obtain the wave equation for the pressure:

$$\frac{1}{c^2} \frac{\partial^2 p}{\partial t^2} + \frac{2Y\rho_0}{r} \frac{\partial p}{\partial t} = \frac{\partial^2 p}{\partial x^2} + \frac{\partial p}{\partial x} (\ln S)_x'. \quad (3)$$

For pressure varying harmonically in time, $p(x, t) = p(x)e^{-i\omega t}$, Eq. (3) takes the form of the generalized Webster equation

$$p'' + p'(\ln S)' + p\left(k_0^2 + \frac{2Y\rho_0}{r}(i\omega)\right) = 0, \quad (4)$$

where $k_0 \equiv \omega/c$ and the prime denotes the derivative with respect to x . For zero-valued admittance Y , Eq. (4) transforms to the conventional Webster equation.

ACOUSTIC BLACK HOLE. THE WKB APPROXIMATION

For the sake of definiteness, we consider in what follows a particular structure of the waveguide with a varying cross section and varying wall admittance (Fig. 2). This structure is a segment of a perfectly hard cylindrical tube of radius R , inside which a number of perfectly hard rings are mounted rather close to each other. The inner radius $r = r(x)$ of the rings decreases according to a law satisfying the condition $\lim_{x \rightarrow -0} r(x) = 0$.

The whole structure is located in the halfspace $x < 0$, and its right-hand wall lies in the plane $x = 0$. In this structure, the waveguide is the region AOC, while the remaining part of the tube between the rings forms the varying admittance of the walls of this waveguide. The admittance is determined by the compressibility of the medium between the adjacent rings. Simple calculations give

$$Y = (-i\omega) \frac{1}{\rho_0 c^2} \frac{R^2 - r^2}{2r}. \quad (5)$$

Substituting expression (5) in Eq. (4) and taking into account the relationship $S = \pi r^2$, we obtain

$$p'' + 2p'(\ln r)' + k_0^2 \frac{R^2}{r^2} p = 0. \quad (6)$$

Note that this equation holds for a tube of varying radius R as well.

Following the cited papers [2, 3], we consider the solutions to Eq. (4) in the WKB approximation and determine the functional behavior $r(x)$ that can ensure the infinite time of propagation along a finite waveguide length. Note that the applicability of the WKB approximation assumes that waves propagating in different directions do not transform into each other. Substituting $p(x) \sim \exp(ikx)$ in Eq. (6), we obtain the following expression for the local wave number $k = k(x)$:

$$k(x) = \sqrt{k_0^2 \frac{R^2}{r^2} - ((\ln r)')^2} = \frac{1}{r} \sqrt{k_0^2 R^2 - (r')^2}. \quad (7)$$

The WKB approximation is applicable if coefficients in Eq. (6) vary only slightly within distances about $k(x)^{-1}$:

$$\left| \frac{d(\ln r)'}{dx} \frac{1}{k} \right| \ll |(\ln r)'| \quad \text{and} \quad \left| \frac{d(\ln r)'}{dx} \frac{1}{k} \right| \ll \frac{1}{r^2}.$$

These inequalities can be rewritten in the form

$$|k| \gg \left| \frac{r''}{r'} \right|, \quad |k| \gg \left| \frac{r'}{r} \right|. \quad (8)$$

Consider now the power-law function of x for the waveguide radius:

$$r(x) = \varepsilon x^n. \quad (9)$$

In this case, both conditions (8) are identical,

$$|k| \gg 1/x, \quad (10)$$

and the formula for the current wave number (7) takes on the form

$$k(x) = \frac{1}{\varepsilon x^n} \sqrt{k_0^2 R^2 - \varepsilon^2 n^2 x^{2(n-1)}}. \quad (11)$$

From relations (10) and (11), we obtain the following condition of applicability of the WKB approximation:

$$\left| \frac{1}{\varepsilon x^{n-1}} \sqrt{k_0^2 R^2 - \varepsilon^2 n^2 x^{2(n-1)}} \right| \gg 1.$$

For $n > 1$, this condition holds for sufficiently small

$$|x| \left(|x| \ll \left[\frac{k_0 R}{\varepsilon \sqrt{n^2 + 1}} \right]^{1/(n-1)} \right), \quad \text{and, for } n = 1 \text{ and } \varepsilon \ll$$

$k_0 R$, it holds for any x . Below, we consider the case $n = 1$, when the waveguide radius decreases most slowly and the condition of applicability of the WKB

approximation holds uniformly in x . We choose the parameter ε as the ratio of the radius of the tube to the length L of the segment along which the waveguide radius decreases from R to zero (see Fig. 2): $\varepsilon = R/L$. From Eq. (11), we obtain

$$k(x) = -\frac{1}{x} \sqrt{k_0^2 L^2 - 1}. \quad (12)$$

The minus sign corresponds to the wave propagating toward increasing x , i.e., from left to right. For $x \rightarrow -0$, the wave number tends to infinity. In this case, both phase c_{ph} and group c_g velocities tend to zero,

$$c_{ph} = \frac{\omega}{k} = -\frac{\omega}{\sqrt{k_0^2 L^2 - 1}} x, \quad c_g = -\frac{c^2 \sqrt{k_0^2 L^2 - 1}}{L^2 \omega} x. \quad (13)$$

The propagation time T of a wave packet from the initial cross section $x = -L$ to cross section $x_1 = -l$ is

$$T = \int_{-L}^{x_1} \frac{dx}{c_g} = \frac{L^2 \omega}{c^2 \sqrt{k_0^2 L^2 - 1}} \ln\left(\frac{L}{l}\right). \quad (14)$$

For $x_1 \rightarrow -0$, we have $T \rightarrow -\infty$. A wave incident on the left-hand cross section of the waveguide will never reach its right-hand cross section and, consequently, will never be reflected from the right-hand cross section. Thus, the structure under study is an acoustic black hole similar to the vibrational black hole considered in [2, 3] for bending waves in a bar of varying cross section.

EXACT SOLUTION

It turns out that Eq. (6) with $r(x) = -\frac{L}{R}x$ has a set of exact particular solutions in the form of power-law functions

$$p(x) = A_\alpha \left(\frac{x}{L}\right)^\alpha. \quad (15)$$

Indeed, after substituting Eq. (15) into Eq. (6), the power-law functions of x are canceled, and we obtain a quadratic equation in the exponent α , whose two solutions are

$$\alpha_{1,2} = -\frac{1}{2} \pm \sqrt{\frac{1}{4} - (k_0 L)^2}. \quad (16)$$

For $k_0 L < \frac{1}{2}$, these exponents are real-valued quantities, and the pressure field exhibits behavior of a non-wave type (pressures at different points are not shifted in phase relative to each other). For $k_0 L > \frac{1}{2}$, the exponents have an imaginary part, and the pressure field demonstrates the behavior of a traveling wave (a finite

phase shift is present). The correspondence of exact solutions (15) and (16) to the WKB approximation is obvious. The power-law function on the right-hand side of Eq. (15) can be rewritten as

$$x^\alpha \equiv x^{-1/2} \exp\left(\pm i \int_L^x \sqrt{k_0^2 L^2 - (1/4)} \frac{dx}{x}\right). \quad (17)$$

Comparing this formula with expression (12) for the local wave number in the WKB approximation, one can see that Eq. (12) correctly describes the functional dependence on the parameter $k_0 L$, except that the root argument has the term $-1/4$ in exact solution (17) instead of the term -1 in Eq. (12). Thus, the threshold frequency, i.e., the frequency starting from which the pressure field demonstrates the wave-type behavior, appears to be two times lower than the value predicted in the WKB approximation. The factor of the exponential in Eq. (17) describes the conservation of the energy flux.

SOUND REFLECTION FROM THE BLACK HOLE

Consider the sound reflection from the black hole and calculate its acoustic admittance. To take into account possible imperfection of the waveguide structure, we assume that the inner radius of rings decreases to a certain finite value, rather than to zero. Thus, the retarding structure occupies the region $-L < x < -l$, where $l > 0$. A unit-amplitude acoustic wave is incident on this structure from the half-infinite homogeneous tube $x < -L$ and is reflected with a coefficient V (see Fig. 3). Two waves are generated inside the black hole. They are described by relationship (15) and have the amplitudes A_1 and A_2 . We specify the admittance Y in the cross section $x = -l$ (at the output of the black hole) and calculate the reflection coefficient V . At the input of the black hole ($x = -L$), the boundary conditions have the form

$$e^{-ikL} + V e^{ikL} = A_1 + A_2, \quad (18)$$

$$\frac{1}{\rho_0 c} (e^{-ikL} - V e^{ikL}) = -\frac{1}{i \rho_0 \omega L} (\alpha_1 A_1 + \alpha_2 A_2). \quad (19)$$

At $x = -l$, the boundary condition is

$$\begin{aligned} A_1 \left(\frac{l}{L}\right)^{\alpha_1} + A_2 \left(\frac{l}{L}\right)^{\alpha_2} &= \frac{V}{Y} \\ &= -\frac{1}{i \omega \rho_0 l Y} \left(\alpha_1 \left(\frac{l}{L}\right)^{\alpha_1} A_1 + \alpha_2 \left(\frac{l}{L}\right)^{\alpha_2} A_2 \right). \end{aligned} \quad (20)$$

Solving Eqs. (18)–(20) simultaneously, we find the reflection coefficient V :

$$V = \frac{1 + W + \frac{c}{i \omega L} (\alpha_1 + W \alpha_2)}{1 + W - \frac{c}{i \omega L} (\alpha_1 + W \alpha_2)} e^{-2ikL}. \quad (21)$$

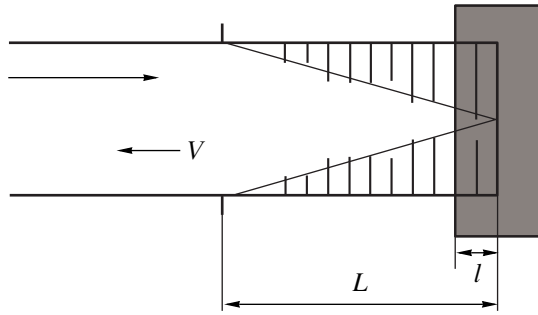


Fig. 3. Reflection of sound from an actual acoustic black hole. The distance l characterizes the imperfection of the black hole.

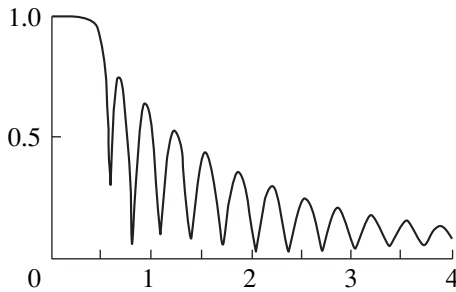


Fig. 5. Frequency dependence of the reflection coefficient of sound from an acoustic black hole in the case of a reflecting (perfectly hard) cross section at $x = -l$ with consideration for the energy loss in the course of the sound propagation in a liquid.

Here,

$$W \equiv \frac{A_2}{A_1} = -\left(\frac{l}{L}\right)^{\alpha_1 - \alpha_2} \frac{1 + \frac{\alpha_1}{i\omega\rho_0 l Y}}{1 + \frac{\alpha_2}{i\omega\rho_0 l Y}} \quad (22)$$

is the coefficient of reflection from the output cross section $x = -l$.

The acoustic admittance of the structure shown in Fig. 3 is expressed as follows:

$$Y_{Bh} = \frac{1}{\rho_0 c} \frac{e^{-ikL} - V e^{ikL}}{e^{-ikL} + V e^{ikL}} = \frac{1}{i\rho_0 \omega L} \frac{\alpha_1 + W\alpha_2}{1 + W}. \quad (23)$$

If the waveguide is made so that no reflection occurs from the cross section $x = -l$, i.e., W is zero, then, we have

$$V = \frac{1 + \frac{\alpha_1 c}{i\omega L}}{1 - \frac{\alpha_1 c}{i\omega L}} e^{-2ikL}. \quad (24)$$

Figure 4 shows the magnitude of the reflection coefficient $|V|$ as a function of ω . As expected, a sharp decrease can be seen in the reflection coefficient at $k_0 L = 0.5$ [compare with Eq. (16)]. The fact that this curve is

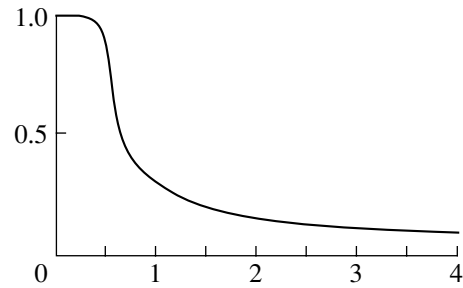


Fig. 4. Frequency dependence of the reflection coefficient of sound from an acoustic black hole in the case of a nonreflecting cross section at $x = -l$ (the frequency is given in $k_0 L$ units).

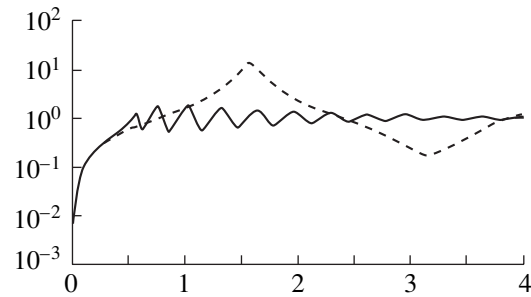


Fig. 6. Acoustic admittance of the black hole (the solid line) in comparison with the quarter-wavelength tube segment (the dashed line). The ordinate axis uses a logarithmic scale.

smoothed and does not exhibit stepwise behavior [which seemingly should appear in accordance with Eq. (16)] is explained by the reflection from the input cross section $x = -L$.

Now, let us assume that $W \neq 0$, $l = 5 \times 10^{-5}$, $L = 0.5$, and the boundary $x = -l$ is hard. We introduce a small energy loss by adding an imaginary part to the velocity of sound: $c = 340(1 - 0.05i)$. Figure 5 shows the corresponding magnitude of the reflection coefficient $|V|$ as a function of ω . The oscillations of the reflection coefficient are related to the resonance features of the black hole. Figure 6 shows the admittance of the black hole characterized by the same parameters as those used for Fig. 5 (the solid line) and the admittance of the standard quarter-wavelength tube segment (the dashed line). Both curves are normalized by $(\rho c)^{-1}$. First and foremost, the curves show that the retarding structure has many resonance peaks, the first resonance frequency being lower than the resonance frequency of the quarter-wavelength tube by a factor of about 3. Thus, the retarding structure increases the effective length of the tube.

With decreasing loss, the oscillation amplitude of $|Y_{Bh}|$ sharply increases. Figure 7 shows the same admittances $|Y_{Bh}|$ and $|Y_0|$ obtained for the sound velocity c

with the imaginary part decreased by four orders of magnitude.

PROPAGATION OF SHEAR WAVES

Note that a similar effect can appear in the case of a shear wave propagation in a medium with varying shear modulus.

To explain the experiments related to acoustic wave propagation in a shallow sea, a model was proposed [4] for a bottom composed of a sediment layer in which the shear modulus decreased as the bottom–water interface was approached. The shear modulus μ was described by the power-law function with the exponent 2ν :

$$\mu = \rho c_0^2 \left(\frac{z}{H} \right)^{2\nu}. \quad (25)$$

Here, c_0 is the velocity of a shear wave at $z = H$ and ρ is the sediment density (assumed to be constant). The shear wave propagation was investigated in a layer with this kind of depth-dependent shear modulus by assuming that the boundary conditions corresponded to the contact with a nonviscous liquid at $z = 0$ and to a hard boundary at $z = H$.

The wave equation for shear waves has the form

$$u'' + \frac{1}{\mu} \mu' u' + \frac{\rho \omega^2}{\mu} u = 0, \quad (26)$$

where the prime denotes the derivative with respect to z . The exact solution given in [4] for an arbitrary ν is expressed through the Bessel functions:

$$u = z^{1/2-\nu} \left[B_1 J_m \left(\frac{\omega}{c_0(1-\nu)} \left(\frac{z}{H} \right)^{1-\nu} \right) + B_2 Y_m \left(\frac{\omega}{c_0(1-\nu)} \left(\frac{z}{H} \right)^{1-\nu} \right) \right],$$

$$m = 2\nu - 1/[2(1-\nu)].$$

At $\nu = 1$, the index m of the Bessel functions is infinite; for this reason, this case was not considered in [4]. Let us consider this special, but interesting, case corresponding to a shear modulus varying with depth according to the quadratic law in the sediment layer:

$$\mu = \rho c_0^2 (z/H)^2. \quad (27)$$

In this case, wave equation (26) takes on a form similar to Eq. (6):

$$u'' + 2 \frac{u'}{z} + \frac{\omega^2}{c_0^2} \left(\frac{H}{z} \right)^2 u = 0. \quad (28)$$

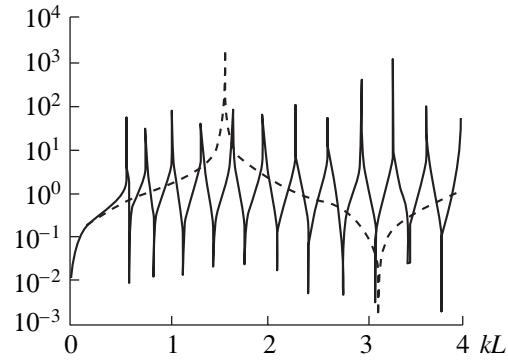


Fig. 7. Same as in Fig. 6 for a sound energy loss in liquid that is four orders of magnitude smaller.

Performing calculations similar to those described above for Eq. (6), we obtain an expression for the local wave number $k = k(z)$ in the WKB approximation:

$$k(z) = \frac{i}{z} \pm \frac{1}{z} \sqrt{\left(\frac{\omega}{c_0} \right)^2 H^2 - 1}. \quad (29)$$

The group and phase velocities of shear waves tend to zero when z tends to zero (i.e., when the sediment layer–water interface is approached), and the time T of the wave packet propagation from the initial cross section $z = H$ to the cross section $z = z_1$,

$$T = \left| \int_{z_1}^H \frac{dx}{c_g} \right| = \frac{H^2 \omega}{c_0^2 \sqrt{(\omega/c_0)^2 H^2 - 1}} \ln \left(\frac{L}{z_1} \right), \quad (30)$$

tends to infinity when $z_1 \rightarrow 0$. A wave propagating from the bottom toward the sediment–water interface never reaches this interface. Thus, we obtain an effect similar to that considered above, i.e., the acoustic black hole. In this case ($\nu = 1$), the wave equation (28) for shear waves in the sediment layer also has a set of exact particular solutions in the form of power-law functions:

$$u(z) = A_\alpha (z/H)^\alpha, \quad (31)$$

$$\alpha_{1,2} = -\frac{1}{2} \pm \sqrt{\frac{1}{4} - \frac{\omega^2}{c_0^2} H^2}. \quad (32)$$

When the exponents are real, i.e., when $\omega < c_0/(2H)$, the field $u(z)$ exhibits no wave-type behavior, whereas, in the opposite case $\omega > c_0/(2H)$, the field given by Eq. (31) is a traveling wave.

CONCLUSIONS

1. We showed conceptually that there is a possibility of creating structures ensuring the total absorption of acoustic waves with the use of nonabsorbing materials only. The absorption occurs as a result of a gradual decrease in the velocity of sound along the waveguide

and the accumulation of acoustic energy at the point where the velocity vanishes.

2. In the special case corresponding to the uniform WKB asymptotics for such a retarding structure with varying parameters, exact solutions are described by power-law functions.

3. If an acoustic black hole is truncated, the retarding structure causes a sharp decrease in the resonance frequencies.

4. A similar effect can be observed for shear waves propagating in a medium with a varying shear modulus.

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